Accuracy of Rise Velocity Estimation from Lagrangian Particle Tracking Data:
A Comparative Analysis of Scattered and Interpolated Approaches

A systematic numerical analysis has been conducted to assess the accuracy of bubble rise velocity estimation from Lagrangian particle tracking data, using a 2D axisymmetric rising bubble at Re = 100, Bo = 100 as the test case. The analysis considered two rise velocity definitions (centroid-based and volume-averaged), two data representations (scattered and re-interpolated), and four interpolation scheme combinations for the gas fraction and velocity fields.

The centroid-based rise velocity consistently outperforms the volume-averaged method, with RMS errors that are generally smaller. This superior accuracy is attributed to the centroid method's exclusive dependence on the gas fraction field, which is interpolated more faithfully than the velocity field during particle tracking.

Re-interpolation of scattered Lagrangian data onto a fixed Eulerian grid dramatically improves centroid-based estimates but offers limited benefit for the volume-averaged method, whose accuracy is fundamentally constrained by velocity field interpolation errors.

The dominant source of error in the Lagrangian framework is the velocity field interpolation, as confirmed by direct comparison of gas fraction and velocity profiles along diagnostic cut-lines. This observation motivates the use of higher-order interpolation schemes (such as fifth-order Lagrangian polynomials) for the velocity field in future tracking simulations, which is currently being investigated.

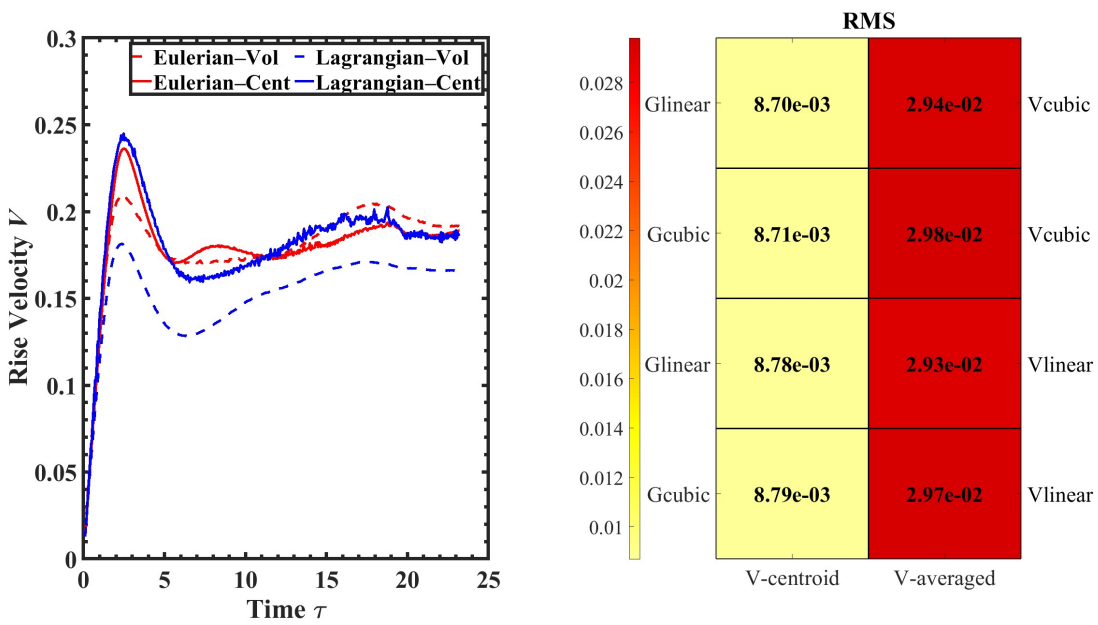

Figure 1: Scattered Lagrangian data. **Left:** Temporal evolution of the non-dimensional rise velocity, comparing the Eulerian reference (red) with the scattered Lagrangian estimate (blue) for the case with linear interpolation scheme for velocity and gas volume fraction. Solid lines correspond to the centroid-based method; dashed lines to the volume-averaged method. **Right:** RMS error heatmap across all four interpolation combinations. Rows indicate the gas fraction interpolation scheme and columns distinguish between centroid-based and volume-averaged methods. Labels on the right margin identify the velocity interpolation scheme.

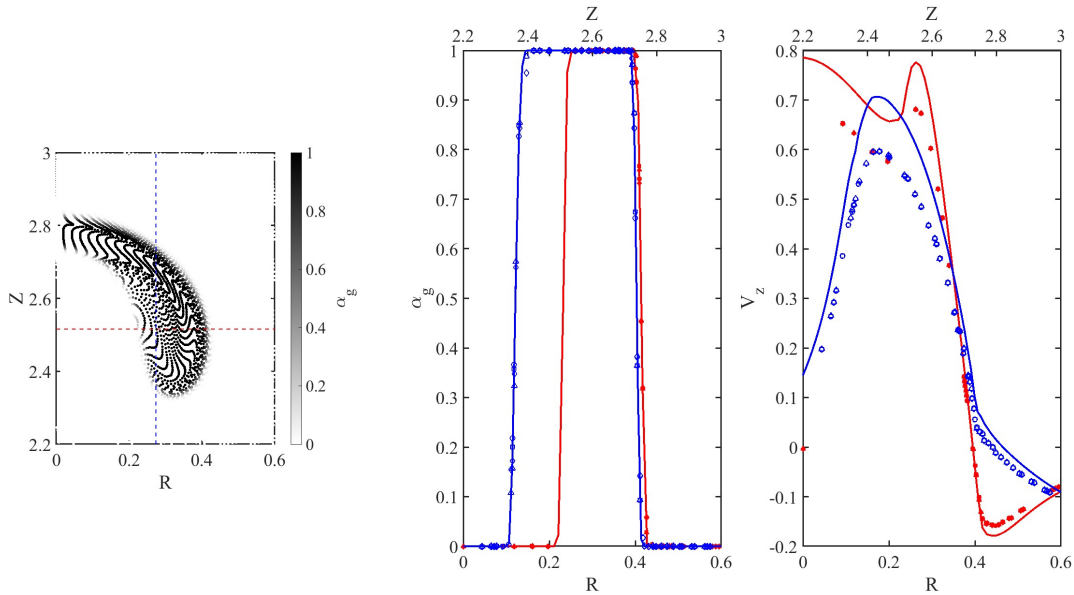

Figure 2: Scattered Lagrangian data at $\tau \approx 6.3$. **Left:** Scatter plot of Lagrangian particle positions coloured by the gas volume fraction α_g; dashed lines indicate the horizontal (red, r-direction) and vertical (blue, z-direction) cut-lines. **Middle:** Gas fraction profiles along the two cut-lines: Eulerian reference (solid lines) versus scattered Lagrangian values (markers) for all four interpolation combinations. Red denotes the r-direction (bottom axis); blue denotes the z-direction (top axis). **Right:** Corresponding axial velocity profiles. The larger scatter in velocity compared with gas fraction confirms that V_z interpolation is the dominant error source.

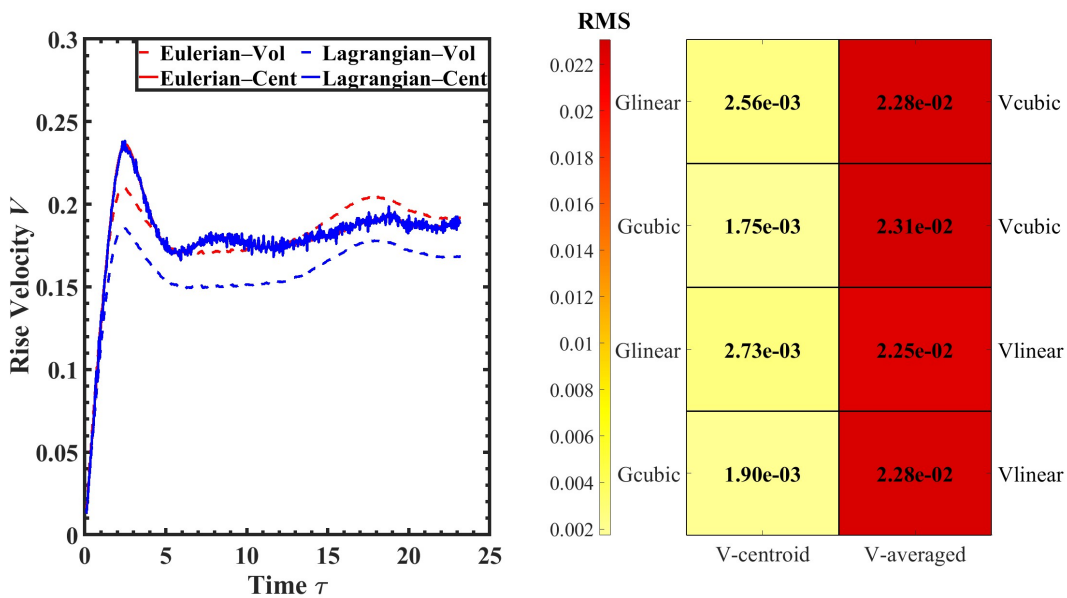

Figure 3: Re-interpolated Lagrangian data. **Left:** Temporal evolution of the non-dimensional rise velocity after re-interpolation. The centroid-based Lagrangian estimate (blue solid) now nearly overlaps the Eulerian reference (red solid), while the volume-averaged estimate (blue dashed) remains offset. **Right:** RMS error heatmap. Centroid-based errors have decreased by a factor of 3–5 compared with fig. 1; volume-averaged errors show only modest improvement.

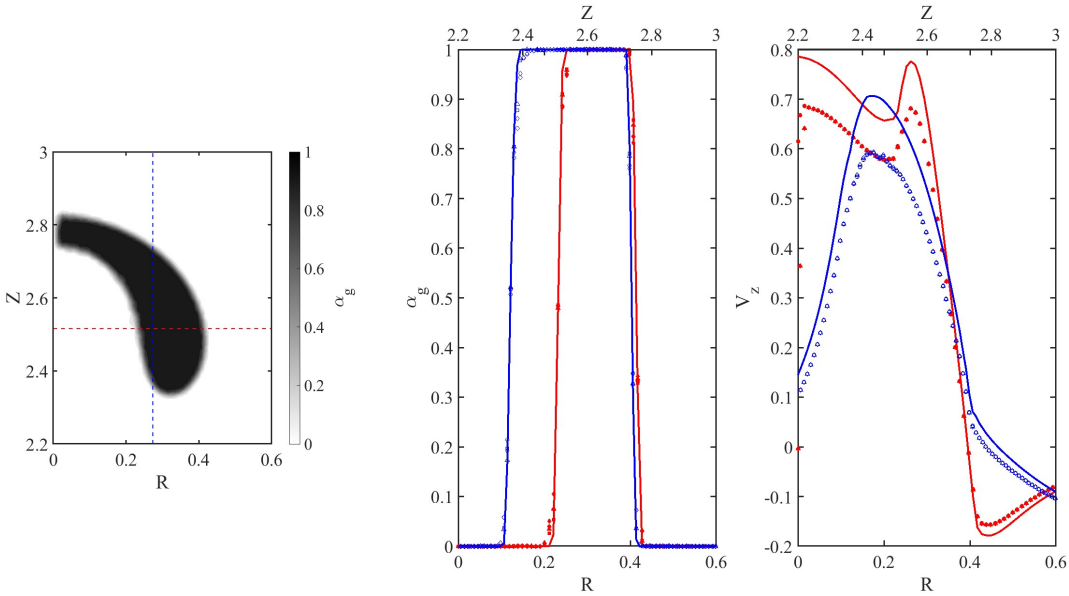

Figure 4: Re-interpolated Lagrangian data at $\tau \approx 6.3$. **Left:** Contour plot of the re-interpolated gas fraction field for the first Lagrangian case, with diagnostic cut-lines indicated. **Middle:** Re-interpolated gas fraction profiles along the cut-lines, now smooth and closely matching the Eulerian reference. **Right:** Re-interpolated velocity profiles. While smoother than the scattered counterpart (fig. 2), deviations from the Eulerian reference persist, particularly inside the bubble.